

VRDots Document Index

Last updated: 2026-04-07

Unity Experiment Assets (Assets/ExperimentSpecs/)

All assets: 63 dots/field, 3.5° aperture radius, 2.0m view distance, 75 Hz, 80ms translation, 750ms delayed onset, 300ms pre-translation hold, 81°/s rotation, 2.26°/s translation.

Asset	depthSep	delayTrans	linkDepthColor	Swaps	Notes
Exp_Baseline	—	1	—	None	No-swap baseline; balances R/G color
Exp_MotionSwap	—	1	—	Motion	100% coherence swap at tStart; balances R/G
Exp_Dots50Swap	—	1	—	Dots50	50% dot swap (half switch fields); balances R/G
Exp_DepthBaseline	0.10m	1	—	None	Depth planes but no swap
Exp_DepthParam_003m	0.03m	1	—	None	Parametric depth, R/G balanced, no swap
Exp_DepthParam_005m	0.05m	1	—	None	Parametric depth, R/G balanced, no swap
Exp_DepthParam_010m	0.10m	1	—	None	Parametric depth, R/G balanced, no swap
Exp_DepthParam_015m	0.15m	1	—	None	Parametric depth, R/G balanced, no swap
Exp_DepthSwapCtrl	0.05m	1	—	N/ZdA/ZdB	All-red (no color); ZdA=cued translator changes plane, ZdB=non-cued changes
Exp_DepthColorLinked	0.05m	1	1	ZdA/ZdB	Depth+color always co-occur (linked); no N baseline
Exp_DecoupledDots_005m	0.05m	1	0	N/C/Z/CZ	PRIMARY. delayTranslator=1 (delayed field translates = CUED); R/G balanced
Exp_DecoupledDots_Inv_005m	0.05m	0	0	N/C/Z/CZ	PRIMARY (inverted). delayTranslator=0 (always-on field delayed); CUED/UNCUED labels inverted before analysis

SwapType codes: **N**=nothing, **C**=color only, **Z**=depth only, **CZ**=color+depth; **ZdA**=cued translator depth changes, **ZdB**=non-cued depth changes.

Data Sessions (chronological)

Session	Asset used	N valid	Key result
260323_1534	Baseline	~128	+50.0pp cueing
260324_0716	Baseline	~128	+28.1pp (cursor-jump issues)
260324_1010	MotionSwap	~128	+27→15pp; motion swap reduces cueing

Session	Asset used	N valid	Key result
260325_1039	Dots50Swap	~128	+34pp; 50% swap does NOT reduce cueing
260325_1831	DepthBaseline 0.10m	~64	+27.5pp overall; Far=+59pp, Near=-5pp
260325_1914	DepthBaseline 0.10m	~64	+8.6pp; Far=+65pp, Near=-47pp
260325_2013	DepthBaseline 0.03m	~64	+16.6pp; depth barely perceptible
260326_1649	DepthSwap50_005m	~128	+40.3pp; Zd=+36pp, N=+45pp
260330_1853	DepthSwapCtrl_005m	192	+34pp binocular; N=+34, ZdA=+12 n.s., ZdB=+56*
260330_2012	DepthSwapCtrl_005m	192	+12pp monocular (R eye); ZdA/ZdB pattern attenuated
260331_0621	DepthSwapCtrl_005m	192	Binocular session 2
260331_1530	DepthSwapCtrl_005m	192	Monocular R-eye #2
260331_1705	DepthSwapCtrl_005m	192	Monocular L-eye #1
260331_1734	DepthSwapCtrl_005m	192	Monocular L-eye #2
260402_0624	DepthParam_010m	128	+47pp; Far=+47, Near=-22
260402_0656	DepthParam_015m	128	+56pp; Far=+47, Near=-25
260402_0716	DepthParam_003m	128	+30pp; Far=+47, Near=+13
260402_0757	DepthParam_005m	128	+19pp; Far=+47, Near=-9
260404_0940	DepthColorLinked_005m	256	+19pp; ZdNoi=+47, ZdCoh=+19
260404_1123	DepthColorLinked_005m	256	+21pp; flatter, UNCUEd elevated
260406_1532	DecoupledDots_005m	514	S1; normal labels
260406_1754	DecoupledDots_Inv_005m	512	S2; labels inverted
260407_0643	DecoupledDots_Inv_005m	512	S3; labels inverted
260407_0731	DecoupledDots_005m	513	S4; anomalous (flat +5pp n.s., elevated UNCUEd, included)

Key Results

DecoupledDots GLM — S1+S2 combined (n=1026, primary)

Factor	LPM	Odds Ratio	p
F1 Dot cueing (temporal onset)	+22.3pp	OR=3.07	*
F2 Depth-field cueing (translator in correct depth plane)	+12.5pp	OR=1.89	*
F3 Color-field cueing (translator matches delayed field color)	+0.0pp	OR=1.00	n.s.

Baseline (UNCUEd+CZ) = 12.5% ≈ chance. OR is the appropriate effect size (baseline near floor).

DecoupledDots — All 4 sessions combined (n=2051)

Swap	CUED	UNCUEd	Δ	sig
N	48.4%	24.9%	+23.5pp	*
C	50.0%	26.5%	+23.5pp	*
Z	23.4%	14.4%	+9.0pp	**
CZ	25.0%	19.5%	+5.5pp	†

F1 dot cueing: Δ =+15.4pp χ^2 =59.01 *. **F2 depth-field: Δ =+9-12pp**. F3 color: null.

Full 2×4 χ^2 (7)=148.28 p<10⁻¹⁰■■.

Figures (Agents/Figures/)

DecoupledDots — Per-session (S1–S4)

File	Contents
decoupled_dots_260406_1532.png	S1 (n=514): dot/depth/color 3-factor panels
decoupled_dots_260406_1754.png	S2 (n=512, inverted): 3-factor panels
decoupled_dots_260407_0643.png	S3 (n=512, inverted): 3-factor panels
decoupled_dots_260407_0731.png	S4 (n=513, anomalous): 3-factor panels; note flat dot cueing

DecoupledDots — Combined

File	Contents
decoupled_dots_combined.png	S1+S2 combined (n=1026): all 3-factor panels + summary — primary analysis figure
decoupled_dots_combined_s1s2s3s4.png	All 4 sessions (n=2051): 3-factor combined
decoupled_dots_session_comparison.png	Side-by-side per-session 3-factor effects; shows S4 anomaly flagged with ■
decoupled_dots_glm.png	GLM results: observed vs predicted + coefficient bar chart (dot/depth/color ORs)
decoupled_dots_traj.png	4×4 grid of frame-by-frame motion trajectories for all N/C/Z/CZ × CUED/UNCUED conditions

DecoupledDots — Factor performance (new 2026-04-07)

File	Contents
decoupled_dots_factor_performance.png/.pdf	6-panel figure (all 4 sessions, n=2051). Row 1: cued vs uncued bar for each factor (A=dot, B=depth, C=color). Row 2: pairwise cued-arm comparisons (D=dot vs depth, E=dot vs color, F=depth vs color). Wilson 95% CI + χ^2 brackets.
decoupled_dots_dot_vs_depth.png/.pdf	2-panel figure . Panel A: 2×2 matrix (Dot✓Depth✓/Dot✓Depth✗/Dot✗Depth✓/Dot✗Depth✗) — direct head-to-head, shows dot wins. Panel B: all 8 raw conditions (CUED/UNCUED × N/C/Z/CZ) with depth-swap cost annotations vs N baseline.
decoupled_dots_report.pdf	5-page comprehensive report . P1: 3-factor Δ overview + per-session table. P2: cued vs uncued % correct for each factor. P3: all 8 conditions by swap. P4: 2×2 conflict + 8-condition disruptiveness. P5: pairwise cued-arm comparisons with interpretation.
decoupled_dots_field_cueing.png	Earlier standalone field-cueing figure (superseded by combined scripts)

DecoupledDots — Dot trace visualizations (new 2026-04-07)

File	Contents
dot_trace_decoupled_N_CUED.png	Single aperture view: N/CUED/Near/heading=0°. Shows accumulated dot positions ± 6 frames around translation. Filled=Near, Open=Far, Red=FieldB, Green=FieldA, alpha gradient pre→trans→post.
dot_trace_decoupled_multi_near.png	2×4 grid: CUED/UNCUED rows × N/C/Z/CZ cols, delayed field=Near. Shows depth and color swaps at tStart as filled↔open and color changes mid-trace.
dot_trace_decoupled_multi_far.png	Same 2×4 grid, delayed field=Far.

DepthSwapCtrl

File	Contents
depthswapctrl_all_sessions_bars.png	Multi-session bars for N/ZdA/ZdB including binocular + monocular sessions
three_factors_bino_vs_mono.png	F1/F2/F3 effects compared: binocular (n=384) vs all monocular pooled (n=769)
trans_plane_2x2.png	2×2 of translating depth plane (Near/Far) × condition (N/ZdA/ZdB)
trans_plane_color.png	Color-coded translating depth plane breakdown

DepthColorLinked

File	Contents
depth_color_linked_results.png	S1+S2 combined results: cueing by ZdNoi/ZdCoh + 3-factor panel
depth_color_linked_traj.png	Frame-by-frame trajectories with R/G depth-color coding
depth_color_biases.png	Bias analysis for color-balanced conditions

DepthParam

File	Contents
depthparam_by_trans_depth.png	Key figure: cueing effect by translating depth plane (Near/Far) across 0.03/0.05/0.10/0.15m — shows Far saturated, Near crosses zero
depthparam_trajectories.png	Frame-by-frame trajectories; Fig 1B convention (green dashed=delayed FieldB, red solid=non-delayed FieldA)

Earlier / General

File	Contents
all_experiments_summary.png	Cross-experiment summary bar chart
baseline_writeup.png	Baseline cueing figure (pre-depth era)
swap_traj_4subfield.png	4-subfield trajectory diagram for motion/dots swap conditions
factor_labeled_trajectories.png	12-panel factor-labeled trajectories (N/ZdA/ZdB × CUED/UNCUED × Near/Far)
feature_trajectories_overview.png	Overview of all swap feature trajectories
flat_permutations.png	Permutation diagram for balanced swap design
labeling_comparison.png	Old vs new Near/Far labeling convention comparison
depth_color_swap_traj.png	Trajectory diagram for depth+color linked swap

WriteUps (Agents/WriteUps/)

File	Contents
baseline_writeup.pdf	Early cueing effect before depth conditions
swap_writeup.pdf	Motion swap (100% reduces cueing) + Dots50 swap (50% does not)
depth_baseline_writeup.pdf	First depth experiments; Far>Near asymmetry emerges
depth_swapctrl_writeup.pdf	DepthSwapCtrl full results: ZdA kills cueing, ZdB enhances; binocular vs monocular
depthparam_writeup.pdf	Parametric depth separation (0.03–0.15m): Far saturated, Near slope
depth_color_linked_writeup.pdf	DepthColorLinked: apparent color effect shown to be depth confound in DecoupledDots
depth_color_linked_conditions.pdf	Full enumeration of all conditions in DepthColorLinked asset
depth_color_design.pdf	Rationale for decoupling color from depth in DecoupledDots
depth_color_decoupled_design.pdf	Full design spec for DecoupledDots (N/C/Z/CZ × CUED/UNCUED orthogonal factorial)
depth_color_decoupled_traj.pdf	Trajectory diagrams for all N/C/Z/CZ conditions
decoupled_three_conds.pdf	Early 3-condition analysis (before CZ added)
decoupled_four_conds.pdf	Full 4-condition (N/C/Z/CZ) analysis document
decoupled_50pct_analysis.pdf	Why 50% partial swaps are geometrically impossible while maintaining clean depth planes
s1s2_swap_analysis.pdf	Why S1↔S2 full attribute swap is valid but functionally null — confirms design symmetry
red_green_motion_lit.pdf	Red vs green in suprathreshold motion: no known red>green advantage; green photopically brighter on Quest
writeup_index.pdf	Earlier index (partially superseded by this file)

Literature / Theory (Agents/Literature/)

File	Contents
decoupled_dots_results.md	PRIMARY CURRENT RESULTS DOC — full write-up: design, all sessions, 3-factor results, GLM, interpretation, open questions, relation to literature
experiment_status.md	Running log of all sessions and results across all experiments
pilot_results_summary.md	Pilot results summary (early sessions)
pilot_summary.md	Condensed pilot summary
data_musing.md	Exploratory observations and hypotheses
factor_labeled_trajectories.md	Factor assignments (F1/F2/F3) for N/ZdA/ZdB conditions — authoritative reference for DepthSwapCtrl
color_cueing_review.md	Color in exogenous attention literature; context for F3=0 null result
color_model_conjecture.md	Theoretical conjecture on why color-field cueing = 0 in DecoupledDots
vergence_latency_note.md	F2 depth effect is neural disparity, not vergence-mediated (80ms < ~70–85ms vergence latency)
depth_ior_hypothesis.md	DepthParam theory: gradient migration account, 5 predictions, IOR abandoned
programmer_critique_gradient_migration.md	Critique and suggested falsifying tests for gradient migration account
modeling_lit.md	Computational/modeling literature (motion coherence, normalization, object-based attention models)
integrated_review.md	Broad literature review integrating VRDots findings with prior work
theory_doc.md	Running theory document: key constructs, predictions, how findings fit
historical_comparison.md	Comparison to prior literature
depthparam_results.md	DepthParam-specific results notes
catak2022_fixation_methods.md	Fixation methods paper notes
endogenous_color_hillyard.md	Hillyard-lab endogenous color cueing literature notes
endogenous_color_summary_and_design.md	Summary + possible design for endogenous color cueing follow-up
pending_papers.md	Papers to read/integrate
paper_notes/	Directory of individual paper notes
document_index.md	This file

Analysis Scripts (Tools/Analysis/)

Script	Purpose
decoupled_dots_combined_analysis.py	Main analysis: per-session (S1–S4) + S1+S2 combined + S1–S4 combined + session comparison figure
decoupled_dots_glm.py	Logistic regression GLM (3 factors); produces coefficient figure
decoupled_dots_factor_performance.py	6-panel figure: cued vs uncued arms + pairwise cued-arm comparisons (all 4 sessions)
decoupled_dots_dot_vs_depth.py	2-panel figure: 2×2 dot×depth conflict matrix + all 8 conditions with depth-swap costs
decoupled_dots_report.py	5-page comprehensive PDF report with commentary
decoupled_dots_field_cueing.py	Standalone field-cueing analysis (superseded by combined script)
decoupled_dots_traj.py	4×4 frame-by-frame trajectory grid (N/C/Z/CZ × CUED/UNCUED)
plot_dot_traces.py	Circular aperture dot trace visualization for DepthSwapCtrl (N/ZdA/ZdB); pre=6/post=6 default
plot_dot_traces_decoupled.py	Circular aperture dot trace visualization for DecoupledDots (N/C/Z/CZ); single-panel or --multi 2×4 grid; pre=6/post=6 default
depthparam_trajectories.py	DepthParam trajectory figure (Fig 1B convention)
depth_color_linked_analysis.py	DepthColorLinked results analysis
depth_color_linked_traj.py	DepthColorLinked frame-by-frame trajectories with R/G coding
gen_hypothetical_traj.py	Generates hypothetical motion trajectories; reference for motion-kind array logic

Script	Purpose
factor_traj_labeled.py	12-panel factor-labeled trajectories (N/ZdA/ZdB × CUED/UNCUED × Near/Far)